Lab Task- 05

Course Title: Technical Writing and Presentation
Course Code: CSE 429

Submitted to:
Md. Imam Mahdi
Lecturer,
Department of EEE
Metropolitan University, Sylhet.

Submitted by:
Orthi Deb Sithi
ID: 231-115-044
Section: B
Batch: 58th
Department of CSE
Metropolitan University, Sylhet.

Date of Submission: 29th March, 2026

1.Software Used:

The online LaTeX editor **Overleaf** was utilized to write, compile, and manage the multiple files required for this comprehensive report, specifically the main.tex and references.bib files.

2. Description of Commands Used

To construct this academic report, a variety of structural, visual, and referencing LaTeX commands were implemented:

Table Formatting Commands:

1. **`\begin{table} [h]`**: Initiates the table float environment and instructs LaTeX to place it approximately "here" in the document.
2. **`\begin{tabular} {|l|c|c|}`**: Defines the table structure, where `|` creates vertical lines and `l/c` determines if the text is left-aligned or centered.
3. **`\hline`**: Inserts horizontal lines to separate header and data rows.
4. **`&` and `\\`**: The ampersand separates data within columns, while the double backslash indicates the end of a row.

Bibliography and Referencing Commands:

1. **\bibliographystyle {plain}**: Sets the visual style of the bibliography to a standard numerical format.
2. **\bibliography {references}**: Directs LaTeX to pull citation data from the external references.bib file.
3. **\cite{citation_key}**: Used inline to call specific references and generate numbered brackets in the text

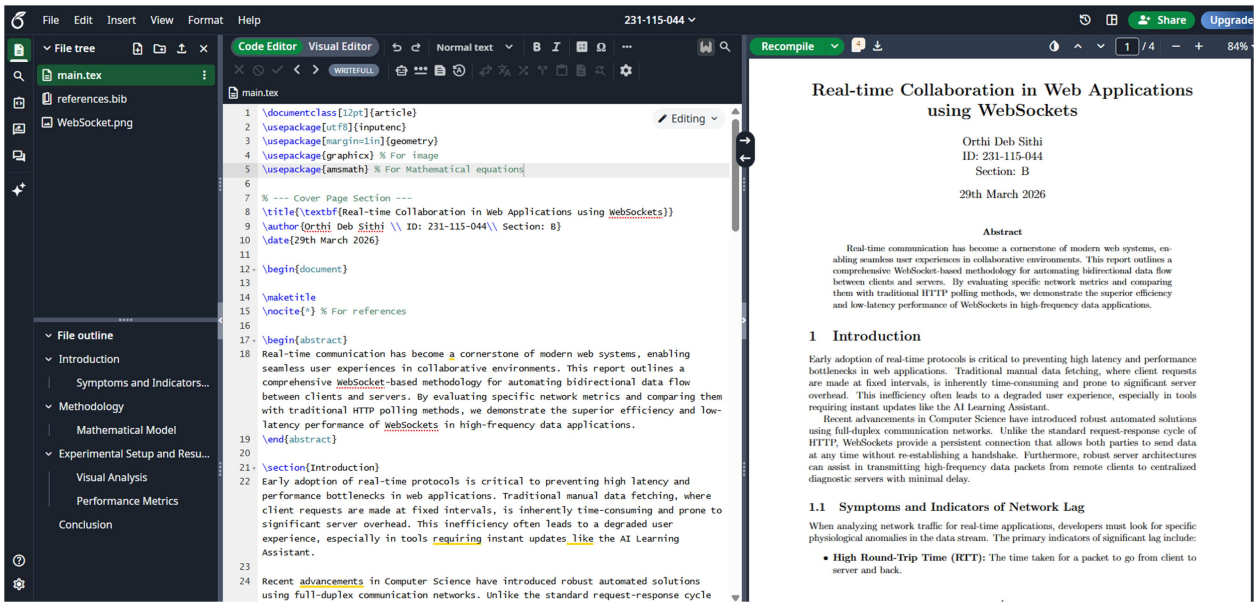
General Structural Commands:

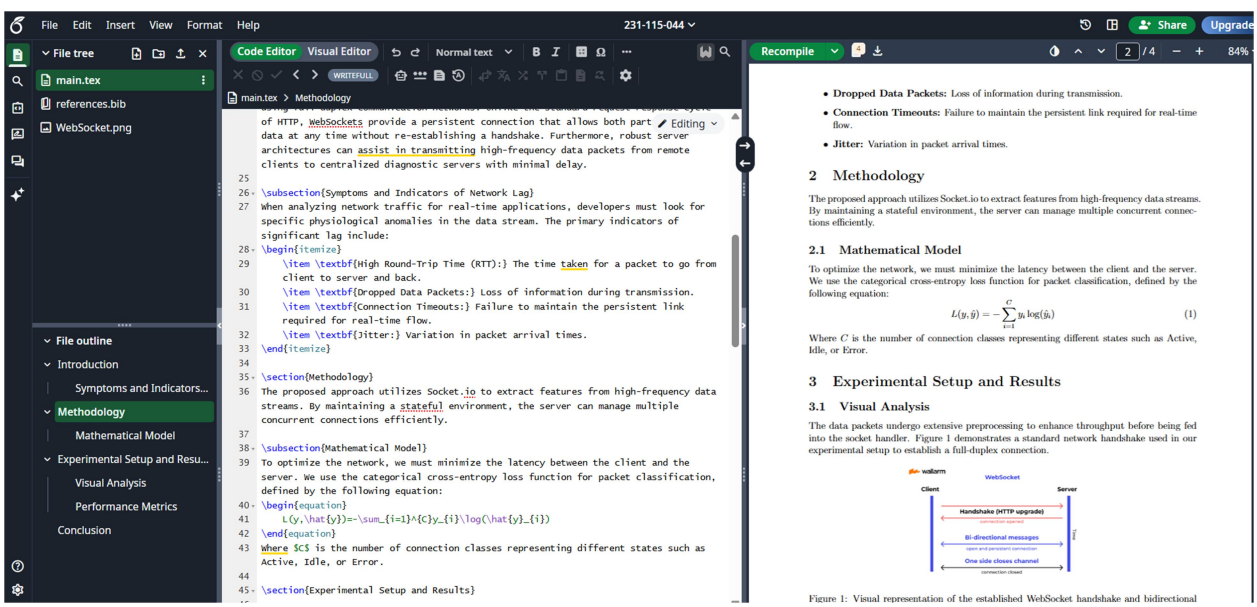
1. **\begin{abstract}**: Creates a formatted summary section at the start of the report.
2. **\section{...} and \subsection {...}**: Used to build the logical hierarchy of the document.
3. **\begin{equation} and \begin{figure}**: Used to insert numbered mathematical models and images with captions and labels for internal referencing.

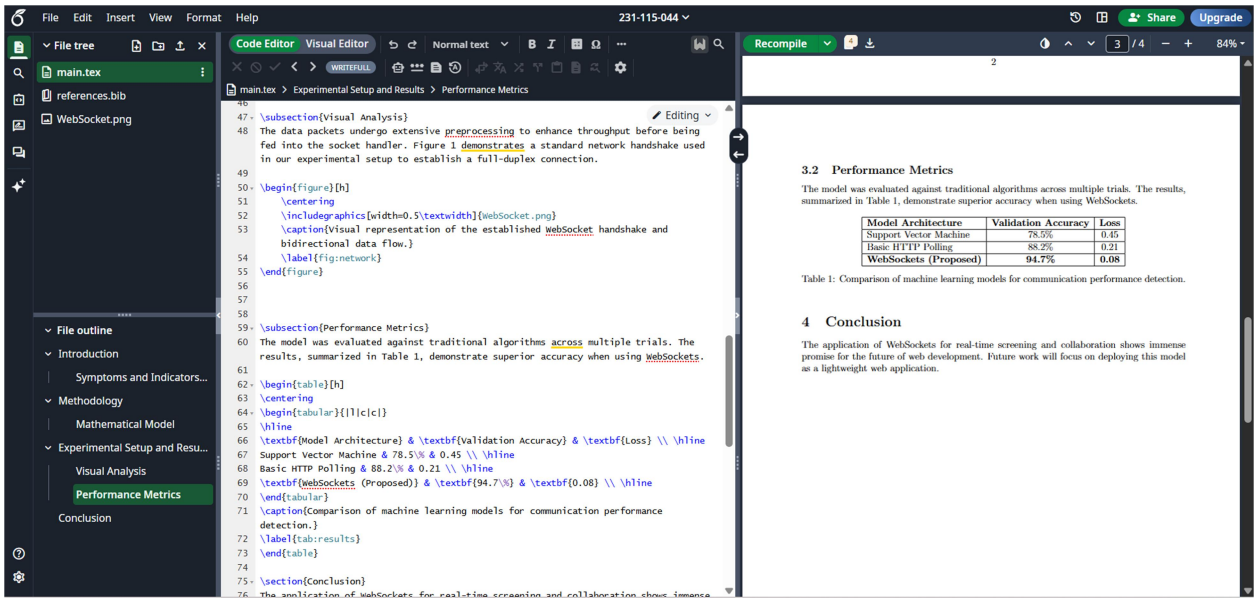
3.Result

Overleaf Project Link:
https://www.overleaf.com/read/ztvccrqxmwkm#4612cf

Code and Output Screenshots








```latex
\section{Conclusion}
The application of WebSockets for real-time screening and collaboration shows immense promise for the future of web development. Future work will focus on deploying this model as a lightweight web application.

\newpage
\bibliographystyle{plain} %
\bibliography{references} %

\end{document}
```

3. Challenges Faced

Several technical challenges were encountered during the compilation of this report:

1. **Bibliography Compilation Errors**: Initially, the reference list would not display. It was discovered that LaTeX requires active \cite{} commands within the body text for references to compile and display correctly.
2. **Table Syntax Complexity**: Formatting required strict attention to detail. Missing a single alignment character (&) or line break (\\) would cause the entire tabular environment to break.
3. **Managing Multiple Files**: Coordinating the main.tex and references.bib files required careful management of citation keys to ensure they matched the BibTeX data exactly.